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Illicit discharge detection in stormwater drains using an Arduino-based low-cost sensor network

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ABSTRACT

Illicit discharges in urban stormwater drains are a major environmental concern that deteriorate downstream waterway health. Conventional detection methods such as stormwater drain visual inspection and dye testing have fundamental drawbacks and limitations which can prevent easy location and elimination of illegal discharges in a catchment. We deployed 22 novel low-cost level, temperature and conductivity sensors across an urban catchment in Melbourne for a year to monitor the distributed drainage network, thereby detecting likely illicit discharges ranging from a transitory flow with less than 10 minutes to persistent flows lasting longer than 20 hours. We discuss rapid deployment methods, real-time data collection and online processing. The ensemble analysis of all dry weather flow data across all sites indicates that: (i) large uncertainties are associated with discharge frequency, duration, and variation in water quality within industrial and residential land uses; (ii) most dry weather discharges are intermittent and transient flows which are difficult to detect and not simply due to cross-connections with the sewerage network; (iii) detectable diurnal discharge patterns can support mitigation efforts, including policies and regulatory measures (e.g., enforcement or education) to protect receiving waterways; and, (iv) that it is possible to cost effectively isolate sources of dry weather pollution using a distributed sensor network.

Key words: distributed sensing, dry weather flow, illicit discharges, low-cost sensing, pollution source tracking, real-time monitoring

HIGHLIGHTS

- Low cost water level, temperature, electrical conductivity sensors deployed at 20+ locations in catchment.
- IoT based sensor server ensures network is affordable, detects dry weather anomalous discharges.
- Most detected discharge pulses last under an hour, making it hard to detect via traditional sensing.

INTRODUCTION

Urban stormwater typically contains high levels of pollutants including sediments, nutrients, heavy metals and faecal bacteria that may degrade water quality in downstream waterbodies (Makepeace *et al.* 1995; Booth & Jackson 1997). Although most studies focused on the characterisation and mitigation of non-point sources originating from urban surfaces during rain events (Duncan 1999; Wong 2006), research has emphasised that urban dry weather pollution discharges from wastewater, industrial and commercial point sources contribute similar if not higher levels of pollution than surface runoff (Pitt & McLean 1986; McPherson *et al.* 2005; Sercu *et al.* 2009; Deffontis *et al.* 2013). Without detecting and regulating these dry weather pollution sources, improvements to downstream water quality will be hampered (Shi *et al.* 2019).

In the context of urban stormwater drainage, dry weather discharges include sewage, wash water (e.g., greywater or car-wash effluent) and liquid waste (e.g., oil and process water). These directly enter the stormwater sewer via cross-connections, misconnections, or indirectly from drainage inlets due to unauthorised dumping, accidental spills, and outdoor washing Brown *et al.* (2004). Dry weather discharges may be hard to detect since they can occur sporadically in urban catchments. Conventional sensory detection methods adopted by Illicit Discharge Detection and Elimination (IDDE) programs include visual inspection for the presence of flow and deposits (Irvine *et al.* 2011), dye testing to locate

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the sources of cross-connections (Schmidt & Spencer 1986), conditioning monitoring via video inspection to reveal misconnections or pipe defects (Ellis & Butler 2015), and dogs trained to identify sources of wastewater (Van De Werfhorst *et al.* 2014). However, these methods have inherent limitations: visual inspection being subjective (Dirksen & Clemens 2008) and rarely able to trace to the actual source of discharge; video inspection being time consuming and ineffective outside peak hours; dye testing requiring access to private properties (Panasiuk *et al.* 2015). Furthermore, the limited availability of field staff and monetary resources also limits the rigour of storm drain inspections.

In addition to detecting the presence of urban dry weather discharges, grab water samples are commonly collected to identify sources of discharge using indicator parameters including temperature, ammonia, conductivity, caffeine, microbial parameters such as *E. coli*, or even detailed DNA fingerprinting to target human wastewater (Irvine *et al.* 2011; Harwood *et al.* 2014; Panasiuk *et al.* 2015; Henry *et al.* 2016). By collectively analysing for such parameters, a quick grab sample can support pollution source tracking. However, the level of detail and temporal resolution of detection achieved by this method is nevertheless limited. Panasiuk *et al.* (2015) highlights four major drawbacks of grab sampling: (1) it remains difficult to detect and capture illicit discharges when they occur, especially for intermittent and transitory flows; (2) poor sampling location – discharge occurring roughly more than 1 km upstream would result in no detection due to dilution (Panasiuk *et al.* 2016); (3) long turnover times to receive analytical results (Schang *et al.* 2016); and (4) high cost of chemical and microbial source tracking analyses.

Recent developments in on-line continuous monitoring techniques provide a much greater opportunity to detect intermittent dry weather flows, and studies have mainly focused on detecting abnormal temperature levels in drains to locate illicit connections, including distributed temperature sensing (Hoes *et al.* 2009) and use of Infra-Red cameras (Lepot *et al.* 2017). Similar to conventional sensing technology such as a submerged flow sensor (HACH Company n.d.), however, the equipment is expensive and cumbersome to install. As such, online methods have mainly been used for research and small-scale studies due to budget limitations of IDDE programs undertaken by municipalities or water authorities. To overcome the existing disadvantages of such online sensor methods and leverage their potential benefits, Shi *et al.* (2021) developed an Arduino-based low-cost sensor that measures water level, temperature, and electrical conductivity. The sensor is easy and relatively cheap to install and connects to a broad range of microcontroller units for real-time data collection – making it substantially more feasible to install sensors throughout stormwater drainage networks to isolate dry weather flows to practical drainage lengths for follow up inspection.

This study aims to develop a systematic approach based on this low-cost sensor technique (Shi *et al.* 2021) to assist with illicit discharge detection and elimination in urban stormwater drains. The objectives of this study include: (1) verify the low-cost sensor's performance in real-time detection of dry weather discharges; (2) understand the characteristics of dry weather discharges including their frequency, duration, and intensity for different land uses and predict potential sources; and (3) evaluate the efficacy of such a spatially distributed sensing network to isolate dry weather discharges and support appropriate management responses.

METHODOLOGY

Study site and distributed sensing locations

A suburban catchment towards the east of Melbourne, Australia with mixed residential and industrial land use was selected as the case study for this research. The 160-ha industrial region of the catchment was previously identified to be a significant source of pollutants to the receiving natural waterway. In total, 22 storm drains including 17 industrial and five residential/commercial sites (see Figure 1) were monitored for more than a year between 3rd July 2019 and 2nd Aug 2020.

Each site's upstream boundary was estimated based on the drainage network and surrounding topography. For site IND03 and IND04 (along the same drainage line), their boundaries were generated by isolating the area of upstream sites (i.e., for IND03, removing the boundary of IND04 and RC05) such that each site had a single predominant land use (i.e., a single land use >30% of area). Site characteristics that were determined for our analysis included: drainage infrastructure dimensions, upstream catchment area, imperviousness, land use proportions and the weighted average age of sewer and drainage infrastructure.

Low-cost real-time logger and sensor

To detect dry weather discharges, two versions of loggers and sensors were trialled in this study. The first version was used for about half of the monitoring period until January 2020 (Figure 1(a)). It includes an Arduino Uno board (an open-source

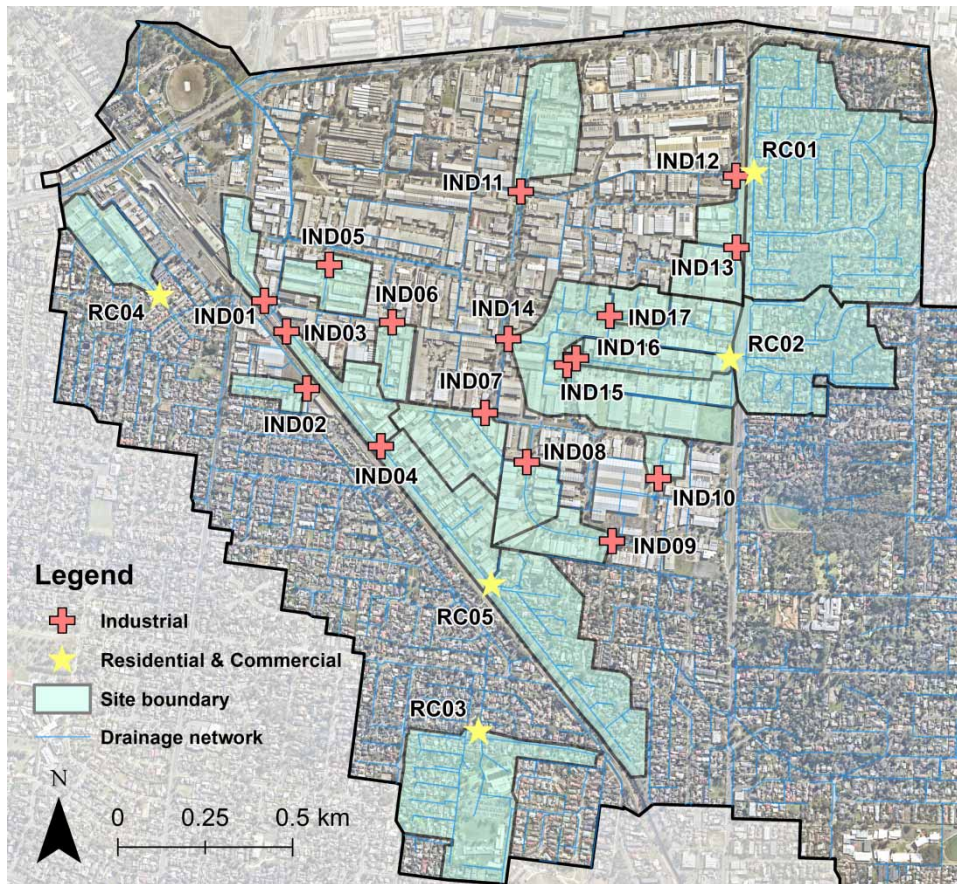


Figure 1 | Sensors' installation location and their estimated sub-catchments (shaded light blue area); Red cross sites: industrial-predominant land use; Yellow star: residential-predominant land use; Blue line: the existing drainage network.

microcontroller for developing digital devices such as data loggers and sensors) to collect data from an integrated water level and temperature sensor (model MS5803 – [TE Connectivity \(2017\)](#)), and measures electrical conductivity (EC) by using an EC sensor modified from a US standard power plug ([Ratcliffe 2015](#)). Collected data were uploaded to a web database in real time via the Arduino-compatible 3G telemetry module SIM5320 ([SIMCom 2018](#)). The first version was powered by a 7.2 V 10.05Ah Lithium-Ion battery with an average lifetime of 2–3 weeks. The second version ([Figure 2\(b\)](#)) utilised an enhanced data logger, the BoSL board, which was specifically designed for real-time water monitoring purposes ([BoSL Water Monitoring and Control 2020](#)). BoSL boards have all the existing functions of an Arduino MCU at reduced power consumption, which achieves a lifetime between six months to one year by using a 3.7 V 11.2Ah LiIon battery. An all-in-one water level, temperature, and EC sensor ([Shi *et al.* 2021](#)) also replaced the separate sensor system used in the first version to reduce its preparation and installation time. The sensor unit including the data logger, the all-in-one sensor, and the battery, together costs around USD 150.

Installation method and site maintenance

For urban drainage monitoring, the standard installation method is through fixing the sensor on the invert of the pipe, thus requiring confined space entry, at least three crew members and involves hours of setup and retrieval. In this study, two rapid installation methods by using a sandbag or a pole were developed and tested for different types of manholes/pits, thus helping to reduce setup and maintenance times and improve safety of personnel.

Sandbag method

This method was originally used on all the sites for the first version of low-cost sensors, and it only took around 20–30 minutes to install. This method is intended to hold a pool of water and slowly release the water downstream. The water pool keeps the

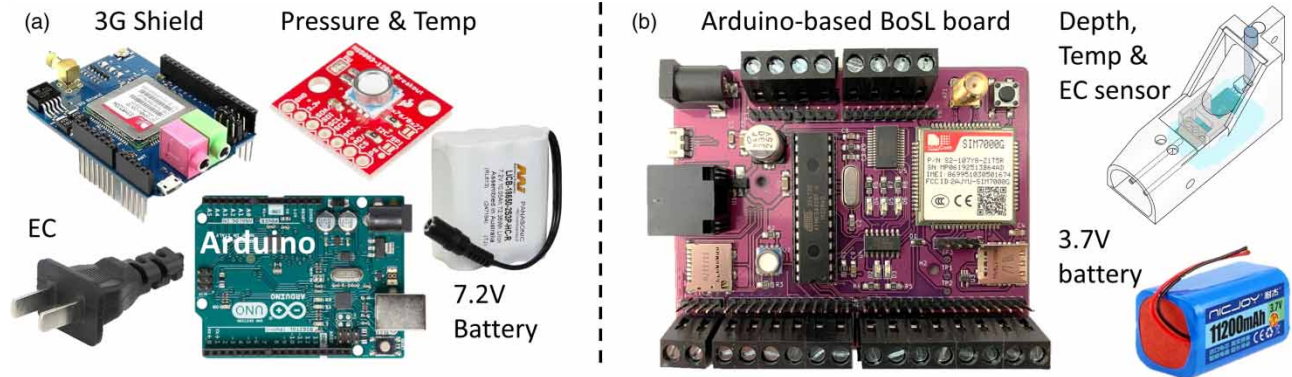


Figure 2 | (a) the trial version of Arduino-based data logger with SIM5320 3G telemetry shield, and two individual sensors including MS5803 pressure and temperature sensor and a US standard power plug-based EC sensor, powered by a 7.2 V Lithium-ion battery; (b) the enhanced version of Arduino-based BoSL board that integrated a SIM7000 NB-IoT telemetry module and powered by a 3.7 V lithium-ion battery; the all-in-one sensor that consists of pressure, temperature and EC sensors in one customised case for easy installation.

EC sensor inundated to provide reliable readings even when the incoming flow level is low. However, this can result in water level readings higher than the actual flow level in the drain. During the dry weather period, the sandbag's weight can successfully hold the sensor in its ideal position, but during large rain events, the sandbag has the potential to be washed towards the side or even completely flipped over, which requires site maintenance after each event to correct the sensor position (Figure 3(a)). This results in poor quality data between an event and its next scheduled maintenance.

Pole method

This method was designed to fix the sensor position issue found with the sandbag method. By attaching the sensor at the bottom end of a steel pole and fixing its top end to the concrete wall with anchors, the sensor's position will not be affected by major flows (Figure 3(b)). It was recommended in this study to retain the sandbag to provide reliable EC readings. By preparing all the required poles and hooks in the lab, installing one sensor into the drain took no more than 15

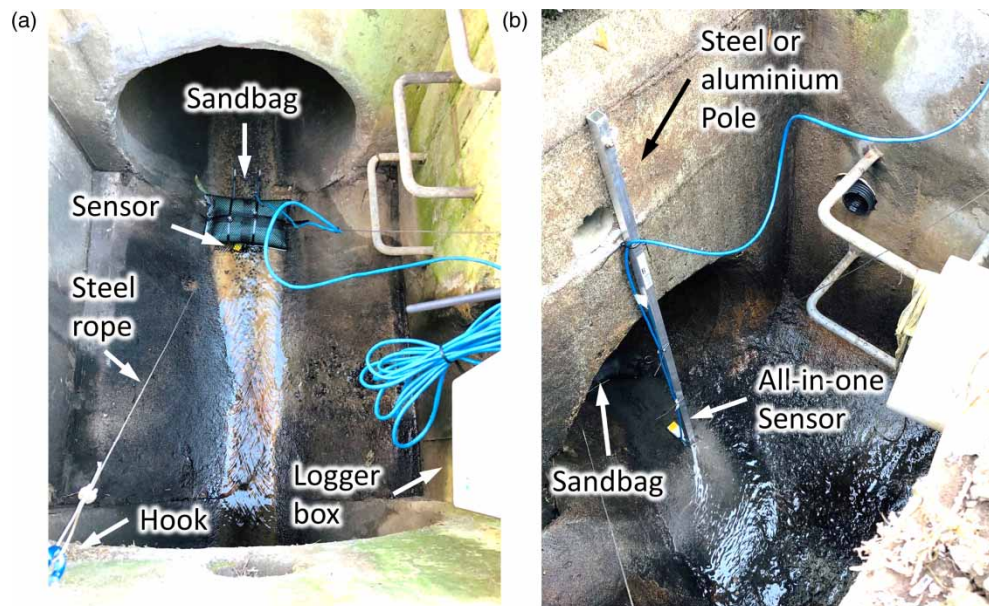


Figure 3 | Both methods have the waterproofed logger box secured inside the pit with no assets on the ground; (a) the sandbag method utilised a customised size sandbag (designed based on the size of the drain); two steel ropes that hang on the hook to protect the sandbag from being washed away; (b) a steel or aluminium pole is used to hold the sensor in position during both dry and wet weather periods.

minutes. In total, 10 sites switched to the pole installation method in November 2019 to January 2020 (RC2, 4, 5 and IND7, 8, 12, 14, 15, 16, 17).

Site maintenance

During each site maintenance, on-site crews manually re-calibrated the depth sensor, replaced batteries, repositioned the sandbag to its ideal location and cleaned all debris and gross pollutants that were trapped by the sandbag or the pole. Maintenance log information including date and time, sensor's position when the pit was opened, sensor's working condition and depth recalibration coefficients were recorded on an inspection form for further data cleaning and analyses. On average, 22 maintenance trips were conducted during the study period for each site (i.e., approximately once every two and half weeks). This maintenance frequency is in line with the low-cost depth sensor's manual recalibration frequency suggested by (Shi *et al.* 2021).

Dry weather flow detection and identification

The sensor cleaning algorithm includes functions to distinguish wet and dry weather period of the dataset and identify discharge pulse during dry periods. The algorithm was tested and validated on five different sites with accurate pulse detection results (i.e., discharge during wet weather periods was not accidentally identified as dry weather pulses), before it was used to automatically process the data from the rest of the sampling sites.

Separating dry and wet periods

Two types of rainfall data, areal rainfall intensity that was estimated based on real-time radar imagery (Bureau of Meteorology 2020) and 6-min interval rainfall data collected at the nearest rain gauge (Melbourne Water 2020), were used to separate dry and wet weather of each site. The dry weather period was triggered based on the following two criteria: (1) the total rainfall of the previous 48 hours was smaller than 1 mm based on both radar and rain gauge data (3.4 mm was used for the rainfall estimated by the radar as it was uncalibrated and on average 3.4 times higher than tipping bucket rain gauge measurement); and (2) rainfall in the previous 2 hours was smaller than 0.5 mm. The radar imagery data enables the web server to undertake data processing in real time, which is crucial for the future real-time monitoring of urban water network and sending alarms to the asset manager. However, as the radar data was accidentally missed due to system routine maintenance, for the purpose of this study, rain gauge data was only used to avoid incorrect classification of dry and wet weather periods.

Detecting dry weather discharge pulses

For dry weather periods, three types of flow conditions (no discharge, continuous flow, or pulses) can be observed at each site. When the EC readings were equal to zero, the site was classified as having no discharge. When the depth readings were stable without having an observed increase or decrease (i.e., slope between two consecutive measurements were smaller than ± 0.3 mm/min), it was considered to have continuous dry weather flow in the drain. A dry weather discharge pulse would be identified when: (1) the water depth at the beginning of the pulse was higher than the average of previous 20 measurements and the slope was higher than 0.3 mm/min; (2) the pulse ended either when the water depth was smaller than 1.3 times the average of the last 20 readings and the slope was negative, a poor-quality data point was detected, or wet weather period began; and (3) the pulse must have a duration between 3 mins and 24 hours with the difference between maximum and minimum water depths greater than 5 mm. These thresholds were determined through an iteration process by testing on the collected data of six individual sites. These values were also set to be conservative to ensure no false positive pulses were identified.

Data analyses of detected illicit discharge

Overall dry weather discharge condition

For each site, the total hours of the detected dry weather period, and hours of three types of flow (pulses, continuous flow and no discharge) were calculated based on the cleaned time series data. The percentage time of each flow type over the dry weather period was then calculated for each site. We also calculated the mean and standard deviation of all types of flows by averaging the result from all sites apart from IND14 to understand the average condition and the uncertainty of dry weather flows in industrial and residential/commercial land uses.

Dry weather discharge pulse's characteristics

Instead of focusing on all types of dry weather flows, this analysis targeted specific dry weather pulses with a spike in water level, which is highly likely caused by dry weather discharges. Each site's dry weather pulse characteristics including frequency, duration, intensity, and level of EC were calculated. The total number of pulses detected per year was estimated by using the number of pulses captured during the dry weather period divided by the available dry weather time and multiplied by 365 days. The duration, average EC and intensity of all pulses were summarised for each site, and the data were plotted in box plots (5th and 95th percentiles with no outliers) for comparison. Flow intensity was calculated by using the maximum water depth of each pulse minus the minimum depth. Additionally, all dry weather pulse data were used to estimate the mean number of pulses per year, duration, intensity and EC of industrial and residential/commercial land uses.

Periodicity of dry weather pulses

For sites with more than 30 dry weather pulses per year, patterns in the frequency of pulses of the day and between days of the week were estimated by calculating the percentage of pulses occurred at a certain time (e.g. 6–9am) and on a certain day (e.g., weekday vs. weekends). The average condition of industrial and residential/commercial land uses was also estimated for future guidance on dry weather discharge detection and elimination. The likelihood of observing dry weather discharges (including dry weather pulses and continuous flow) at a specific hour of a certain weekday (e.g., Wednesday 2–3pm) was determined as a heatmap by using that hour of the day's total detected times of flow and divided by the total available hours of dry period in the same time slot. A heatmap was created for each site to represent the likelihood of observing flowing water in the drain.

RESULTS AND DISCUSSION

Overall dry weather discharge conditions

The proportion of different types of dry weather flows (no discharge, continuous flow, and pulses) is shown for each site (Figure 4). On average, the frequency of dry weather flows in an industrial drain is 62.9% ($\sigma = 23.6\%$), with 3.4% ($\sigma = 2.9\%$) chance of detecting dry weather pulses and 59.5% ($\sigma = 23.1\%$) chance of only observing continuous dry weather flows with stable water depth and EC level. In this study, the five monitored residential and commercial sites exhibited lower frequency of dry weather discharges (mean = 49.3%, $\sigma = 21.9\%$), with 1.6% ($\sigma = 1.0\%$) of flow being classified as dry weather pulses, and 47.7% ($\sigma = 22.8\%$) as continuous discharges. This implies that industrial sites have a higher proportion of dry weather flows.

Dry weather pulses – frequency, duration, intensity, and EC

Frequency

In all three classified flow conditions (no flow, continuous flow, and pulses), this study focus on the pulses and their characteristics since they are most likely caused by human-related polluted discharges. The number of dry weather pulses per year

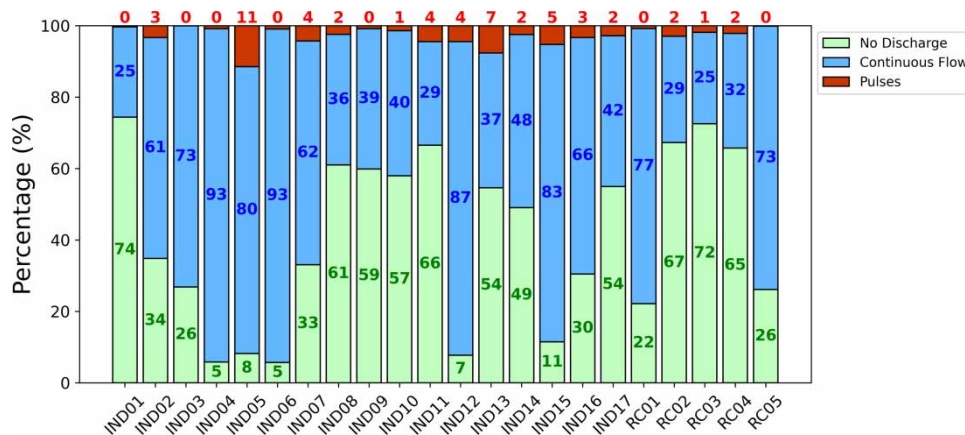


Figure 4 | Percentage time of three types of dry weather flows – no discharge (green), continuous flow (blue) and dry weather pulses (red).

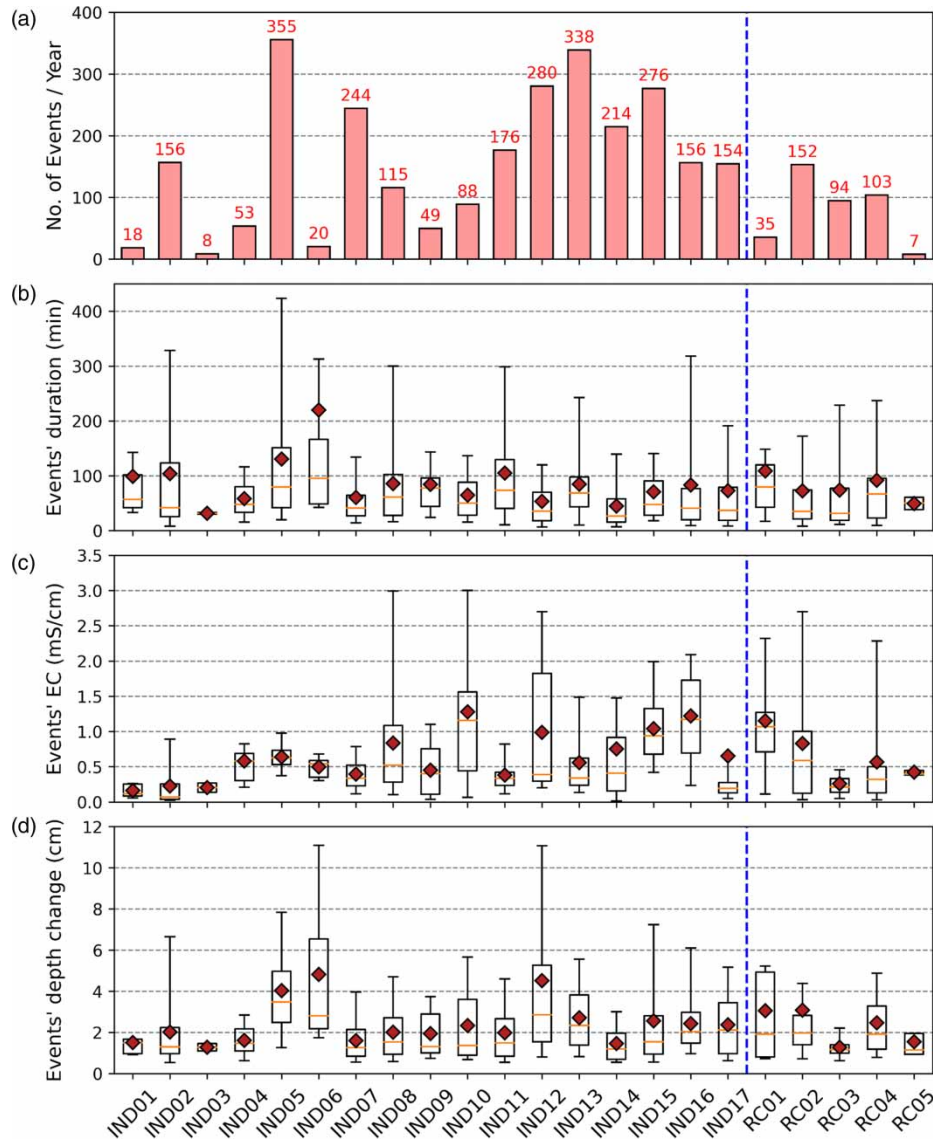


Figure 5 | For each site in this study: (a) number of dry weather pulses detected per year; (b) the box plots of pulses' durations in minutes; (c) the box plots of pulses' EC level in mS/cm; and (d) pulses' flow intensity/flow level increase in box plots.

are shown below in [Figure 5\(a\)](#) for all sites. Industrial sites had an average of 160 pulses/year ranging between eight and 355 detected pulses per year (i.e., almost one pulse per day at IND05). Residential and commercial sites demonstrated an average of 79 dry weather pulses/year ranging from seven to 152 detected pulses per year.

Duration

The duration of dry weather pulses was similar between industrial and residential/commercial sites in this study. Industrial sites had a median flow duration of 51 mins (5th percentile = 9.7 mins and 95th percentile = 270.9 mins). Residential and commercial sites had a median of 45 mins (5th percentile = 8.2 mins and 95th percentile = 281.0 mins). Apart from IND03 and RC05 (having only eight and seven pulses respectively), [Figure 5\(b\)](#) shows that the duration of pulses was positively skewed at all other sites (mean duration: industrial = 83 mins, residential/commercial = 81 mins). This implies that most urban dry weather pulses were intermittent and transient with only a small proportion of continuous discharge that lasted more than 4 hours. Based on previous experience of IDDE, most urban illicit discharges identified in this case would not be detected by conventional detection methods due to the detection window being too narrow ([Brown et al. 2004](#); [Lilly](#)

et al. 2012). However, with the sensors' real-time data capability and opportunity to incorporate alarm triggers through a web interface when a pulse is detected, there is the potential for more rapid pollution response by relevant authorities.

Intensity

Both industrial and residential/commercial sites also had similar flow intensity (Figure 5(c)). The median pulse's intensity was 19.2 mm water depth (5th percentile = 6 mm and 95th percentile = 66.5 mm) and 15.3 mm (5th percentile = 6.6 mm and 95th percentile = 49.1 mm) for industrial and residential/commercial sites, respectively. For most of the study sites, the flow intensity was also positively skewed with more pulses having a smaller water depth. IND05 and IND06 demonstrated the highest flow duration and water depth among all other sites (Figure 5(b) and 5(c)). Regular site maintenance of the sensor found that these two drains were blocked due to a muddy discharge at IND05 and cured concrete in the drainage pipe at IND06. Such a blockage created a 'dam' inside the drain that held the water with higher depth and for a longer period. As such, the irregular duration and flow intensity can be used as a sign of poor-quality dry weather discharges and considered as an early warning of potential pipe blockage.

EC

Figure 5(d) shows that industrial sites showed a median EC level of 0.47 mS/cm (5th percentile = 0.06 mS/cm and 95th percentile = 2.03 mS/cm), which is slightly higher than residential land use with a median conductivity of 0.32 mS/cm (5th percentile = 0.03 mS/cm and 95th percentile = 2.34 mS/cm). Average conductivity levels (industrial – 0.69 mS/cm and residential – 0.61 mS/cm) found in this study are similar to Deffontis *et al.* (2013)'s sampling result at the two catchment outlets.

As discussed by Brown *et al.* (2004), the electrical conductivity of dry weather flows has been considered an important indicator of illicit industrial discharges with a benchmark concentration of 2 mS/cm. In this study, IND08, IND10 and IND16 all had six dry weather pulses with an EC level higher than 2 mS/cm. The industrial activity survey suggested that these three sites all had metal plating and coating factories in operation during the study period. IND16's background (continuous discharge) conductivity was even higher than the EC observed during pulses, implying that the pulse usually had much lower EC levels than the continuous discharge. This suggests that there might be a permanent cross-connection with the industrial wastewater network. IND12 had experienced 31 pulses with conductivity higher than 2 mS/cm, but no properties in the region are known to be associated with metal processing work. A more detailed investigation is needed to track this source of discharge in the catchment. Two residential and commercial sites, RC02 and RC04, also showed eight and five pulses with EC higher than 2 mS/cm, which is roughly five times higher than laundry wash water and 10 times higher than car wash water. However, conductivity was not considered as a key indicator for residential sewage. Further analysis on bacterial tracers such as *E. coli* could help confirm the pollution source (Brown *et al.* 2004).

Dry weather pulses – periodicity

Periodicity between days of the week

To further understand the potential cause of detected pulses, patterns in the frequency between days of the week are plotted in Figure 6. On average, industrial sites had a higher proportion of pulses occurring between Monday to Friday and a lower

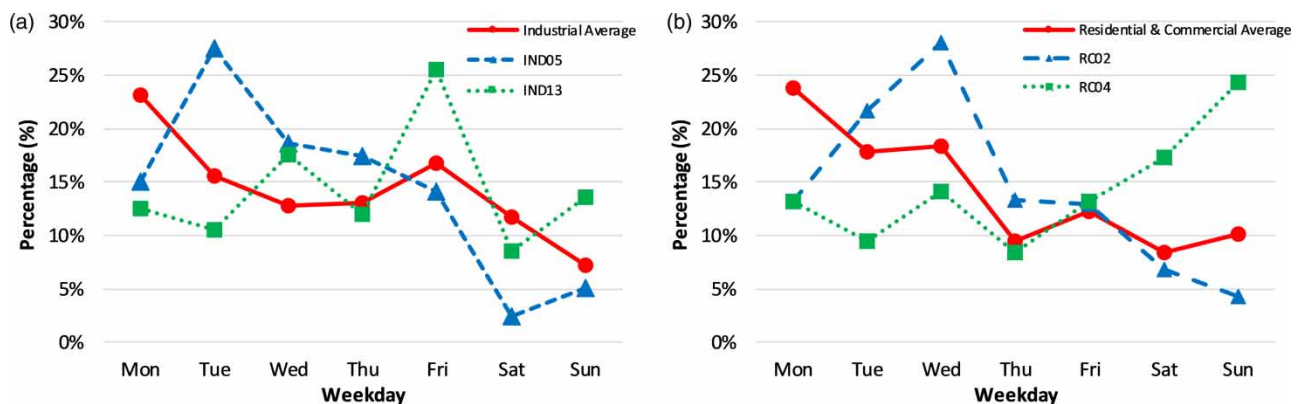


Figure 6 | (a) industrial sites' proportion of dry weather pulses on each weekday (red line) with two other sites that had the highest number of dry weather pulses, IND05 (blue line) and IND13 (green line); (b) residential sites' proportion of dry weather pulses (red line) with two sites that had highest number of pulses, RC02 (blue line) and RC04 (green line).

number of pulses on the weekend (red line – Figure 6(a)), which suggests that the detected dry weather pulses were driven by work activities. Monday had the highest proportion of pulses and Saturday's proportion was close to the level of that of weekdays. Within 13 industrial sites which have more than 30 pulses per year, only three sites, IND02, IND05 (blue line – Figure 6(a)) and IND08, show strong weekday peaks (weekday's daily average pulse proportion is 10% higher than that of the weekend). Another four sites demonstrated that their Saturday's pulse frequency was similar to that of weekdays and significantly higher than Sunday. The remaining six sites including IND13 (green line – Figure 6(a)), had similar pulse frequencies on all seven days of the week (<5%), suggesting that either the discharge's generation site was under operation seven days a week or multiple sources of dry weather discharges may have been present.

Residential and commercial sites in this study tended to have a higher proportion of pulses on Monday, Tuesday, and Wednesday (red line – Figure 6(b)), which diverges from our initial hypothesis that weekends will have a higher number of pulses due to people staying at home or dining out. RC02, a site with 65% residential, 2% commercial and 0% industrial land use demonstrated clear weekday peaks especially on Tuesdays and Wednesdays (blue line – Figure 6(b)), which suggests that take-aways and cafés which mainly served industrial workers could be the source of discharges, instead of residential properties. The only commercial-dominated site in this study, RC04, had a strong weekend peak with 42% of its pulses occurring on Saturdays and Sundays (green line – Figure 6(b)). This might be linked to the higher number of people that run certain businesses in the area that are busier on weekends.

Periodicity within the day

The frequency of pulses occurring at different hours of the day are shown in Figure 7. On average, industrial sites showed a higher proportion of dry weather pulses between 6am and 6pm, which corresponds to general working hours. Around 23% of industrial sites had a clear full-day work hour peak of dry weather pulses (e.g., IND05: blue lines – Figure 7(a)). There were also 46% of the industrial sites, which showed a clear morning or afternoon peak (23% each). What was not expected was that five out of 13 industrial sites exhibited a midnight peak (i.e., between 9pm and 3am) of dry weather discharge pulses (e.g., IND13 with more than 30% of pulses occurring between midnight and 3am: green lines – Figure 7(a)). We discuss IND13 further in the following section.

For residential and commercial sites, the average frequency of pulses between hours of the day were almost evenly distributed between 6am and 3am with a slightly higher proportion of flows between 9am and 9pm (red line – Figure 7(b)). Each residential site also demonstrated its own diurnal pattern. RC04, the commercial-dominated site had most of its pulses occurring between 6am and 6pm, with more than 30% of them in the morning and early afternoon (green line – Figure 7(b)). This may link to businesses such as restaurants in the area. Conversely, the residential-dominated site RC02 had an afternoon and night peak of dry weather pulses. These trends provide valuable information on when to visit the site for a more targeted search to eliminate potential illicit discharges.

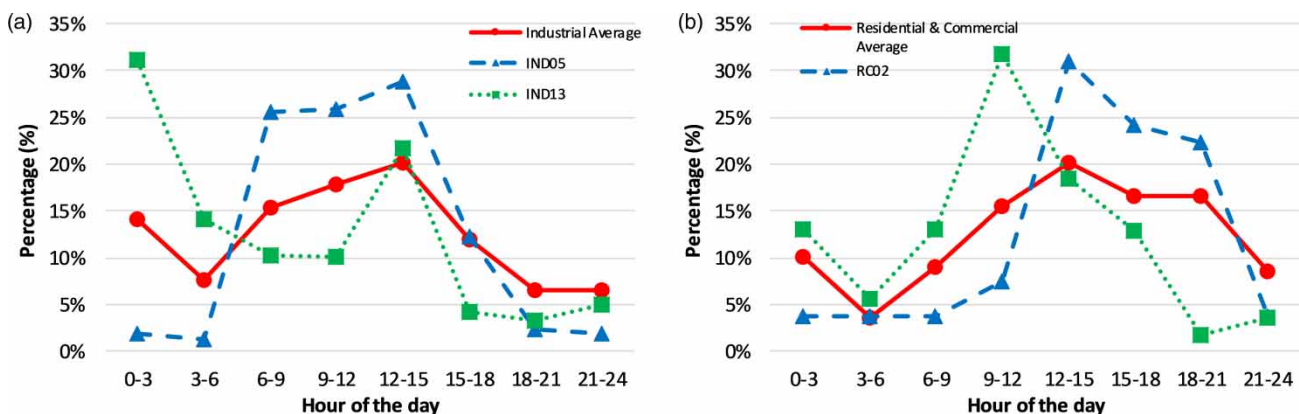


Figure 7 | (a) industrial sites' diurnal pattern of their pulse frequency (red line) with two other individual sites: IND05 (blue line) and IND13 (green line); (b) the diurnal pattern of residential sites' pulse frequency (red line) with two specific sites, residential-dominated RC02 (blue line) and commercial-dominated RC04 (green line).

Daily and hourly periodicity informs when to visit a site

Once dry weather discharges into the urban stormwater drains are identified, source elimination including fixing cross-connections or education programs become essential to stop the improper use of drainage networks. From the low-cost sensor's data, we were able to create a dry weather discharge heatmap (shown in Figure 8) to demonstrate the likelihood of observing any flowing water at every hour during different days of the week. Water utilities and environment protection authorities can use such heatmaps (e.g., generated based on user-defined period of data) as a guide to determine when to visit each site for the best chance of capturing potential illicit discharges, especially for intermittent and transitory flows (dry weather pulses in this study). Overall, the results provide timely and detailed intelligence for regulators to target enforcement actions.

Again, IND05 and IND13, for example, are two sites with confirmed dry weather illicit discharge issues. As shown by Figure 8(a), during the work hour on all weekdays, crews who visited IND05 could have had a 15–40% chance of observing discharges in the drain. The greatest opportunity occurred on Tuesdays between 8am and 1pm where a likelihood of discharge greater than 40% was observed at these two times. Conversely, most of IND13's discharges occurred around midnight making detection by conventional methods impractical as they do not provide long-term real-time monitoring data. Regulating this would be challenging due to occurrence of 'out of work' hours but is potentially achievable by installing one sensor at the legal discharge point of each individual property connected to IND13. Such a sensor could accurately locate the property responsible for the illegally discharged water into this drain at such an unnoticeable time.

Identifying key discharge sources based on the periodicity of dry weather pulses

Unlike conventional dye testing that is conducted at every single property to confirm whether the property has a cross-connection issue, the low-cost sensor method of this study monitors the flow condition in a drainage network at the street scale. This makes the low-cost sensor method able to detect other types of urban dry weather discharges such as illegal dumping and accidental spills.

The number of pulses that were detected per year at each site generally showed the severity of potential unauthorised drainage usage behaviours. By further comparing the periodicity of pulses' frequency with potable water usage patterns, we can potentially confirm the main types of pulses with a site inspection. A cross-connection with a sewer usually results in a morning and evening peak for residential sites, and a workhour peak on workdays for industrial precincts. Other sources of

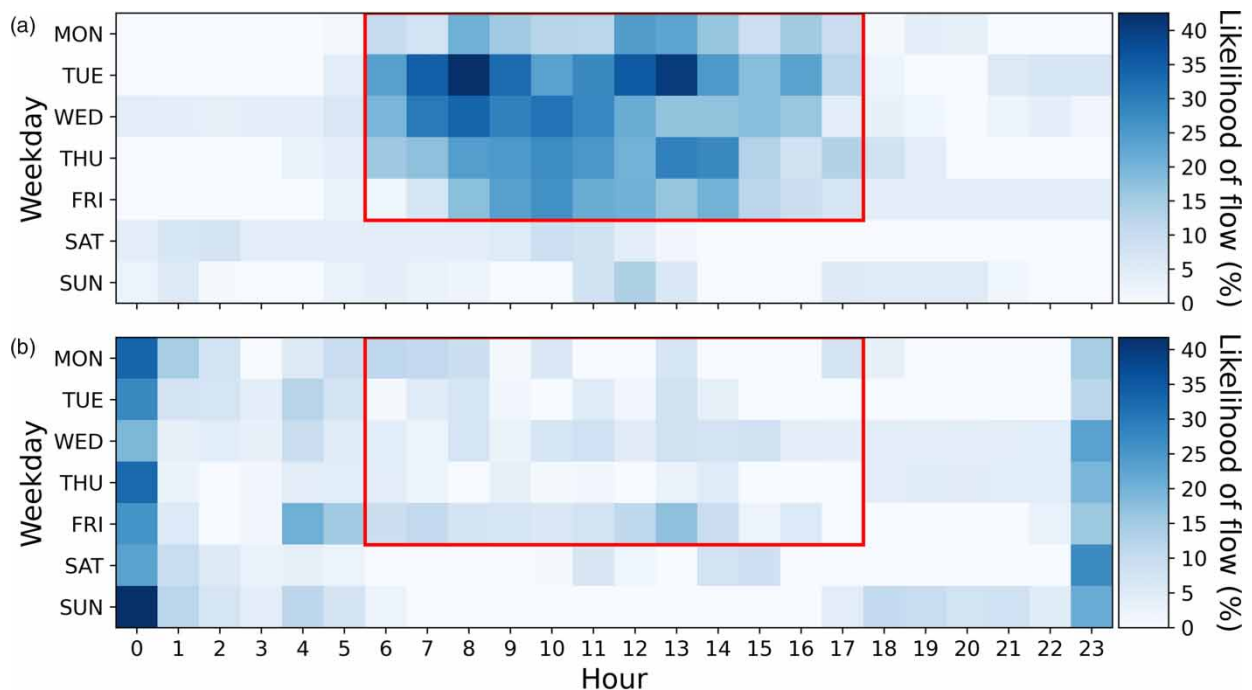


Figure 8 | Dry weather discharge heatmaps that demonstrate the likelihood of observing dry weather flows at a random time of a random day for (a) IND05 and (b) IND13; hours inside the red box represent the workhour of weekdays.

pollution such as surface dumping or washing would be highly related to the specific industrial activities and human behaviours. For example, after the site inspection, the pulse in IND05 was found to be caused by a local mechanic who washed agricultural and construction machinery on weekdays. The prolonged dry weather continuous flow with poor water quality was due to the pipe being partially blocked by the muddy discharge. As such, the illicit discharge entered the local drain through a surface pit rather than the sewer.

CONCLUSIONS

The low-cost distributed sensing approach demonstrated successful detection of dry weather discharges in a stormwater drainage system. Due to its real-time transmission and large volumes of generated data, the sensing network also provides great potential for illicit discharge identification, real-time alarm of pulses that can be responded to quickly, and long-term stormwater surveillance to protect the health of the natural receiving environment.

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DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

REFERENCES

- Booth, D. B. & Jackson, C. R. 1997 [Urbanization of aquatic systems: degradation thresholds, stormwater detection, and the limits of mitigation 1](#). *JAWRA Journal of the American Water Resources Association* **33**, 1077–1090.
- BosL Water Monitoring and Control 2020 *BoSL Board v0.3*. Available from: http://www.bosl.com.au/wiki/BoSL_Board_v0.3 (accessed 11/12 2020).
- Brown, E., Caraco, D. & Pitt, R. 2004 *Illicit Discharge Detection and Elimination: A Guidance Manual for Program Development and Technical Assessments*. Center for Watershed Protection, Ellitcott City, MD, USA.
- Bureau of Meteorology 2020 *64 km Melbourne Radar Loop*. Available from: <http://www.bom.gov.au/products/IDR024.loop.shtml#skip> (accessed 7/10 2019).
- Deffontis, S., Breton, A., Vialle, C., Montrejaud-Vignoles, M., Vignoles, C. & Sablayrolles, C. 2013 [Impact of dry weather discharges on annual pollution from a separate storm sewer in Toulouse, France](#). *Science of the Total Environment* **452–453**, 394–403.
- Dirksen, J. & Clemens, F. 2008 [Probabilistic modeling of sewer deterioration using inspection data](#). *Water Science and Technology* **57**, 1635–1641.
- Duncan, H. P. 1999 *Urban Stormwater Quality: A Statistical Overview*. Cooperative Research Centre for Catchment Hydrology, Melbourne.
- Ellis, J. B. & Butler, D. 2015 [Surface water sewer misconnections in England and Wales: pollution sources and impacts](#). *Science of the Total Environment* **526**, 98–109.
- Hach Company n.d. *Submerged AV Sensor*. Available from: <https://www.hach.com/flow-sensors/submerged-av-sensor/family?productCategoryId=54948588082> (accessed 1 October 2019).
- Harwood, V. J., Staley, C., Badgley, B. D., Borges, K. & Korajkic, A. 2014 [Microbial source tracking markers for detection of fecal contamination in environmental waters: relationships between pathogens and human health outcomes](#). *FEMS Microbiology Reviews* **38**, 1–40.
- Henry, R., Schang, C., Coutts, S., Kolotelo, P., Prosser, T., Crosbie, N., Grant, T., Cottam, D., O'Brien, P. & Deletic, A. 2016 [Into the deep: evaluation of SourceTracker for assessment of faecal contamination of coastal waters](#). *Water Research* **93**, 242–253.
- Hoes, O. A. C., Schilperoort, R. P. S., Luxemburg, W., Clemens, F. & Van De Giesen, N. 2009 [Locating illicit connections in storm water sewers using fiber-optic distributed temperature sensing](#). *Water Research* **43**, 5187–5197.
- Irvine, K., Rossi, M. C., Vermette, S., Bakert, J. & Kleinfelder, K. 2011 [Illicit discharge detection and elimination: low cost options for source identification and trackdown in stormwater systems](#). *Urban Water Journal* **8**, 379–395.
- Lepot, M., Makris, K. F. & Clemens, F. H. 2017 [Detection and quantification of lateral, illicit connections and infiltration in sewers with Infra-Red camera: conclusions after a wide experimental plan](#). *Water Research* **122**, 678–691.
- Lilly, L., Stack, B. P. & Caraco, D. 2012 [Pollution loading from illicit sewage discharges in two mid-Atlantic subwatersheds and implications for nutrient and bacterial total maximum daily loads](#). *Watershed Science Bulletin* **3**, 7–17.
- Makepeace, D. K., Smith, D. W. & Stanley, S. J. 1995 [Urban stormwater quality: summary of contaminant data](#). *Critical Reviews in Environmental Science and Technology* **25**, 93–139.

- Mcpheerson, T. N., Burian, S. J., Stenstrom, M. K., Turin, H., Brown, M. J. & Suffet, I. 2005 Dry and wet weather flow nutrient loads from a Los Angeles watershed 1. *JAWRA Journal of the American Water Resources Association* **41**, 959–969.
- Melbourne water 2020 *Rainfall and River Levels*. Available from: <https://www.melbournewater.com.au/water-data-and-education/rainfall-and-river-levels/> (accessed 11/12 2020).
- Panasiuk, O., Hedstrom, A., Marsalek, J., Ashley, R. M. & Viklander, M. 2015 Contamination of stormwater by wastewater: a review of detection methods. *Journal of Environmental Management* **152**, 241–250.
- Panasiuk, O., Hedström, A., Ashley, R. M. & Viklander, M. 2016 Detection of wastewater discharges into stormwater sewers: effects of travel distance on parameters. *Journal of Environmental Engineering* **142**, 04016016.
- Pitt, R. & Mclean, J. 1986 *Toronto Area Watershed Management Strategy Study: Humber River Pilot Watershed Project*. Ontario Ministry of the Environment, Toronto, Ontario.
- Ratcliffe, M. 2015 *Three Dollar EC – PPM Meter*. Available from: <https://hackaday.io/project/7008-fly-wars-a-hackers-solution-to-world-hunger/log/24646-three-dollar-ec-ppm-meter-arduino> (accessed 2018).
- Schang, C., Henry, R., Kolotelo, P. A., Prosser, T., Crosbie, N., Grant, T., Cottam, D., O'brien, P., Coutts, S. & Deletic, A. 2016 Evaluation of techniques for measuring microbial hazards in bathing waters: a comparative study. *PloS one* **11**, e0155848.
- Schmidt, S. D. & Spencer, D. R. 1986 The magnitude of improper waste discharges in an urban stormwater system. *Journal (Water Pollution Control Federation)* **58**, 744–748.
- Sercu, B., Werfhorst, L. C. V. D., Murray, J. & Holden, P. A. 2009 Storm drains are sources of human fecal pollution during dry weather in three urban southern California watersheds. *Environmental Science & Technology* **43**, 293–298.
- Shi, B., Bach, P. M., Lintern, A., Zhang, K., Coleman, R. A., Metzeling, L., McCarthy, D. T. & Deletic, A. 2019 Understanding spatiotemporal variability of in-stream water quality in urban environments—a case study of Melbourne, Australia. *Journal of Environmental Management* **246**, 203–213.
- Shi, B., Catsamas, S., Kolotelo, P., Wang, M., Lintern, A., Jovanovic, D., Bach, P. M., Deletic, A. & McCarthy, D. T. 2021 A Low-Cost water depth and electrical conductivity sensor for detecting inputs into urban stormwater networks. *Sensors* **21**, 3056.
- Simcom 2018 *SIM5320X Series*. Available from: <https://www.simcom.com/product/SIM5320X.html> (accessed 05/10 2020).
- Te connectivity 2017 *MS5803-01BA Miniature Variometer Module*.
- Van De Werfhorst, L. C., Murray, J. L., Reynolds, S., Reynolds, K. & Holden, P. A. 2014 Canine scent detection and microbial source tracking of human waste contamination in storm drains. *Water Environment Research* **86**, 550–558.
- Wong, T. H. 2006 Water sensitive urban design-the journey thus far. *Australasian Journal of Water Resources* **10**, 213–222.

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